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FRESHAIR DESK

AI-POWERED SMART DESK ENVIRONMENT MONITOR

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BONAFIDE CERTIFICATE

Certified that this project report "FRESHAIR DESK: AI-POWERED SMART DESK ENVIRONMENT MONITOR" is the bonafide work of T. DIVYA DARSHINI (24619232), S. SANTHOSH KUMAR (24619501), and BANGALATHOTA NAVYASREE (24619205), carried out under the supervision of S. M. SOWMIYA, Assistant Professor, Department of Artificial Intelligence and Machine Learning. Certified further, that to the best of our knowledge, the work reported here does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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EXTERNAL EXAMINER

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ABSTRACT

Indoor environmental conditions such as temperature, relative humidity, illumination, and air-quality-related gas readings can change while people study or work at a desk. When these parameters are not observed together, changes may be difficult to notice. FreshAir Desk is an ESP32-based smart desk environment monitoring prototype designed to collect and display these readings through a live laptop dashboard. The prototype uses a DHT22 sensor to measure temperature and humidity, a BH1750 sensor to measure ambient light intensity, and an MQ135 sensor to provide air-quality-related gas-sensing readings. The ESP32 collects the sensor data and passes the readings through a data-processing and rule-based classification workflow. The dashboard presents live values, condition labels, fixed suggestions, and a short history. An AI Insights panel uses the sensor readings to generate a summary, practical suggestions, an overall numeric environment indicator, and a focus-impact statement. These AI-generated outputs are advisory and are not validated measurements of human comfort, health, or productivity. The report describes the system motivation, related work, design, hardware and software requirements, methodology, implementation, observations, limitations, and future enhancements. Comparative scenarios from the project materials are presented descriptively; they are not controlled experiments and do not establish causation or measurement accuracy. The prototype aims to make selected environmental readings visible in one interface and support user awareness of the immediate desk environment.

Keywords: Internet of Things, ESP32, indoor environmental monitoring, smart desk, DHT22, BH1750, MQ135, rule-based classification, AI-generated insights.

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ABBREVIATIONS

Abbreviation	Expansion
AI	Artificial Intelligence
IoT	Internet of Things
RH	Relative Humidity
RS	Sensor resistance
RS/R0	Ratio of sensor resistance to reference resistance
USB	Universal Serial Bus
VOC	Volatile Organic Compound

CHAPTER 1. INTRODUCTION

1.1 OVERVIEW OF THE PROJECT

People spend long periods studying or working at desks, where temperature, humidity, illumination, and air conditions can vary during the day. These parameters are often checked separately or not checked at all. A compact monitoring system can bring several readings together and make the local environment easier to observe.

FreshAir Desk is a smart desk environment monitor built around an ESP32 controller and three sensor modules. The DHT22 measures temperature and relative humidity, the BH1750 measures ambient light intensity, and the MQ135 provides gas-sensing readings associated with air quality. The ESP32 collects and processes these inputs, while a laptop dashboard displays the current measurements and their rule-based interpretation.

The project combines direct sensor observations with two forms of interpretation. The first is a rule-based classification that labels the condition using the configured project logic and provides fixed hints. The second is an AI Insights panel that generates a short explanation and practical suggestions from the available readings. The AI-generated numeric score is a qualitative environment indicator only; it is not a validated comfort score or a direct measurement of concentration or productivity.

1.2 PROBLEM STATEMENT

In many study and work settings, users may not notice changes in lighting, temperature, humidity, or air-quality-related sensor readings until they become uncomfortable or distracting. Using separate devices to check each parameter can be inconvenient, and raw numbers alone may be difficult to interpret.

The project addresses the need for a low-cost, desk-oriented prototype that collects multiple environmental readings and presents them in a single interface. The system

is intended to improve visibility of the measured conditions. It does not automatically control air conditioning, ventilation, or lighting, and it does not claim to diagnose health conditions.

1.3 OBJECTIVES

- To build an ESP32-based prototype for collecting selected indoor environmental parameters.
- To measure temperature and relative humidity using the DHT22 sensor.
- To measure ambient light intensity using the BH1750 sensor.
- To collect MQ135 gas-sensing readings and present them as air-quality-related observations.
- To process the sensor values and display rule-based condition labels on a laptop dashboard.
- To provide fixed condition-linked hints and an AI-generated summary with practical suggestions.
- To document the design, current implementation, observations, and limitations without overstating validation.

1.4 SCOPE OF THE PROJECT

The scope includes sensor interfacing, ESP32-based acquisition, processing of the collected values, rule-based classification, and laptop dashboard presentation. The dashboard displays temperature, humidity, light intensity, MQ135 resistance-related readings, status labels, fixed hints, and a short history. The AI Insights panel generates an advisory summary, suggestions, an overall numeric environment indicator, and a focus-impact statement.

The current project does not include noise sensing, automatic environmental control, a validated comfort score, or a measured productivity outcome. The supplied documentation does not specify every threshold, sampling interval, or the exact

communication protocol between the ESP32 and the laptop. MQ135 values are not documented as calibrated gas concentrations in parts per million.

1.5 APPLICATIONS AND EXPECTED OUTCOMES

Potential applications include student study desks, home workstations, office desks, and small indoor work areas where a user wants to view several environmental parameters together. The intended outcome is a working prototype that collects readings and presents them in a clear dashboard. The system supports awareness; whether it improves health, comfort, or productivity requires separate evaluation.

CHAPTER 2. LITERATURE REVIEW

2.1 LITERATURE OVERVIEW

The supplied literature survey covers IoT-based thermal and indoor-air monitoring, multi-sensor smart-building systems, data-driven indoor air estimation, occupancy inference, and research linking environmental conditions with occupant experience. These studies show that indoor environmental monitoring can be approached as a measurement task or extended toward prediction and control.

Sensor-based systems provide direct observations of selected parameters. More advanced systems may use machine-learning models to estimate air pollution, predict occupancy, or study productivity alongside additional inputs. Such outputs require suitable datasets, clearly defined targets, calibration, and validation. These capabilities cannot be assumed for a prototype simply because it collects environmental readings.

FreshAir Desk follows a narrower monitoring approach. Its confirmed components are the ESP32, DHT22, BH1750, and MQ135, with a laptop dashboard for display. The project's rule-based labels and AI-generated insights are separate from the direct sensor readings. The AI panel provides advisory text, not a validated predictive model for health or productivity.

2.2 COMPARATIVE REVIEW

Study / Theme	**Main Focus**	**Relevance to FreshAir Desk**
IoT thermal and indoor-air monitoring [1]	Real-time observation of indoor conditions	Motivates displaying multiple parameters together.

Study / Theme	Main Focus	Relevance to FreshAir Desk
Air and sound pollution monitoring with HVAC integration [2]	Monitoring linked with building control	Differs because FreshAir Desk does not control HVAC and has no noise sensor.
Machine-learning estimation of indoor air pollution [3]	Predictive estimation using data-driven models	Shows that gas concentration estimation requires a separate model and validation.
Environmental and physiological inputs for productivity prediction [4]	Analysis of productivity-related outcomes	Highlights that productivity claims need dedicated data and evaluation.
Air-quality sensing for headcount estimation [5]	Occupancy inference	A different target from FreshAir Desk's environmental status labels.
IoT smart-building management and occupant prediction [6]	Building-level monitoring and prediction	Provides broader context for connected environmental systems.
Multi-parameter ESP32 monitoring [8]	Microcontroller-based environmental sensing	Offers a hardware-level comparison; its accuracy results cannot be transferred to this prototype.

2.3 SUMMARY OF THE REVIEW

The reviewed literature supports the usefulness of multi-parameter monitoring and shows that environmental data can be used for several different goals. However,

measurement, classification, prediction, and control are distinct functions. FreshAir Desk currently collects selected readings, applies configured rules, and presents advisory insights. No comparative accuracy test against a reference instrument is documented in the supplied materials; therefore, this report does not claim superior accuracy or performance.

CHAPTER 3. SYSTEM DESIGN

3.1 EXISTING SYSTEM

In a basic desk environment, users may rely on personal perception or separate devices to understand temperature, humidity, light, and air conditions. This can make it difficult to see several parameters together. Manual observation also provides limited historical context, especially when conditions change over time.

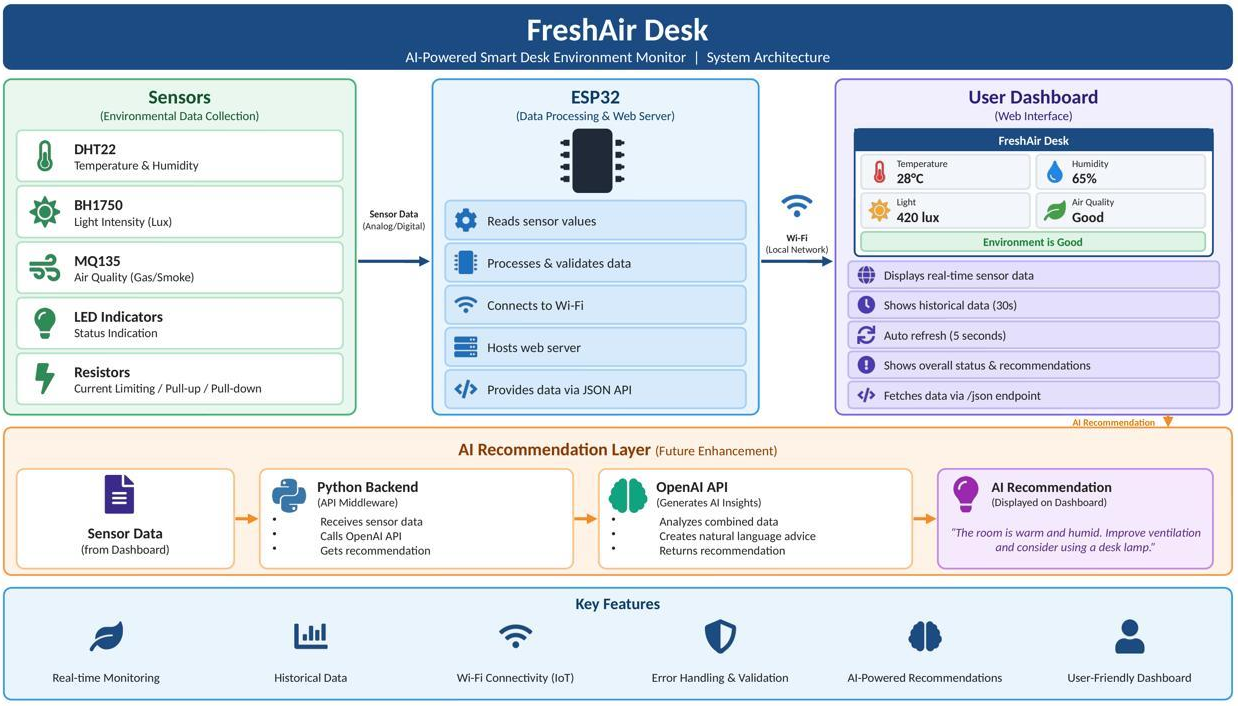
Commercial indoor monitors can provide richer measurements, but device capabilities and cost vary. Some systems also connect to building-management controls or cloud services. FreshAir Desk is designed as a student prototype focused on collecting a small set of readings and presenting them in one local dashboard.

3.2 PROPOSED SYSTEM

The proposed system integrates three sensors with an ESP32 controller. The DHT22 provides temperature and relative humidity, the BH1750 provides illuminance in lux, and the MQ135 supplies gas-sensing resistance-related values. The ESP32 reads the sensors and the software prepares the data for classification and display.

The laptop dashboard displays live sensor values, rule-based labels, fixed suggestions, and a short history. The AI Insights panel separately generates a summary and recommendations based on the readings. The prototype is a monitoring and awareness system; it does not automatically switch appliances or alter room conditions.

3.3 SYSTEM ARCHITECTURE



DATA FLOW: SENSORS → ESP32 → DATA PROCESSING → RULE-BASED CLASSIFICATION → LAPTOP DASHBOARD

3.4 HARDWARE REQUIREMENTS

Component	Purpose
ESP32 Dev Board / Dev Kit V1	Main controller for sensor interfacing, data acquisition, and processing.
DHT22	Measures temperature and relative humidity.
BH1750	Measures ambient illumination in lux.
MQ135	Provides gas-sensing readings associated with air quality.
Breadboard	Supports temporary circuit assembly during prototyping.

Component	Purpose
Jumper wires	Connect the sensor modules to the ESP32.
USB cable	Provides power and supports programming/debugging.

3.5 SOFTWARE REQUIREMENTS

Software / Tool	Purpose
Arduino IDE	Development, compilation, and uploading of ESP32 firmware.
ESP32 board package	Provides ESP32 board support in the Arduino IDE.
Sensor libraries	Support communication with DHT22, BH1750, and associated modules.
Serial monitor / debugging tools	Help observe sensor readings and verify operation.
Dashboard interface	Displays live readings, rule-based labels, fixed hints, and history.
Classification / data-processing logic	Organizes the collected parameters and determines an environmental status.
AI Insights integration	Generates advisory text from sensor readings; output consistency refinements remain ongoing.

3.6 DESIGN CONSIDERATIONS

Sensor readings and interpreted labels are presented as separate types of information. Temperature, humidity, and illuminance are measurements from their respective sensors. MQ135 readings are reported as resistance-related values and are not treated as calibrated ppm concentrations. Rule-based labels depend on the configured project

logic. AI-generated suggestions are advisory and may occasionally be inconsistent with the dashboard status; users should consider the underlying readings rather than treating the generated text as a scientific assessment.

CHAPTER 4. IMPLEMENTATION AND RESULTS

4.1 METHODOLOGY

FreshAir Desk follows a sensor-based IoT methodology. The workflow consists of sensor acquisition, data collection, data processing, environment classification, and user output.

Sensor data acquisition: DHT22, BH1750, and MQ135 collect temperature/humidity, light intensity, and air-quality-related gas readings.

Data collection: the ESP32 reads values from the connected sensor modules.

Data processing: readings are organized and prepared for classification and display.

Environment classification: configured rules interpret the collected parameters and assign the dashboard condition label.

User output: the laptop dashboard presents the current values, status, and associated hints; the AI Insights panel provides a separate advisory summary.

4.2 SENSOR ACQUISITION AND PROCESSING

Sensor / Field	Output	Role in the System
DHT22	Temperature (°C), relative humidity (%)	Shows thermal and moisture conditions near the desk.

Sensor / Field	Output	Role in the System
BH1750	**Illuminance (lux)**	**Shows the amount of ambient light measured by the sensor.**
MQ135	**Sensor resistance (RS) and ratio (RS/R0), where available**	**Provides gas-sensing readings associated with air quality; no calibrated ppm conversion is documented.**
ESP32	**Collected sensor data**	**Reads and organizes values for the dashboard workflow.**

The collected data is passed through the implemented processing path. The exact sampling interval and the communication protocol between the ESP32 and laptop are not specified in the supplied project documentation, so this report does not assign unsupported timing or protocol details.

4.3 CLASSIFICATION AND DASHBOARD

The dashboard provides a consolidated view of the available sensor fields. A rule-based classifier labels the environment as Good, Attention, or Poor and displays fixed, condition-linked suggestions. These labels are produced by project rules and should not be interpreted as medical or regulatory assessments.

A short history helps the user observe recent values rather than relying only on a single reading. The dashboard is intended to make the measured parameters easier to inspect during desk work. The system does not automatically control lighting, air conditioning, or ventilation.

4.4 AI INSIGHTS PANEL

The implemented AI Insights panel receives the sensor readings and generates a summary, practical suggestions, an overall numeric environment indicator, and a focus-impact statement. The panel uses Qwen through Groq, as described in the project materials. The numeric output is intended as a qualitative indication of whether the environment appears poor, acceptable, or very good.

The score is AI-generated and is not a validated comfort score, a clinical measure, or a direct measurement of attention or productivity. Software refinements are ongoing because generated insights may occasionally be inconsistent with the dashboard status. The AI text should therefore be treated as an advisory layer rather than as a substitute for the raw readings or a calibrated instrument.

4.5 OBSERVATIONS AND COMPARATIVE SCENARIOS

The supplied project materials include descriptive comparisons for small and large rooms and classrooms, considering air-conditioned and naturally ventilated conditions and different occupancy levels. These scenarios are useful for organizing observations across indoor settings, but they are not documented as a controlled experiment.

Comparison Dimension	Purpose of Comparison	Interpretation Limit
Room size	Compare a smaller room with a larger room.	Room size alone does not establish the cause of a reading.
Ventilation / cooling	Compare air-conditioned and naturally ventilated scenarios.	Differences cannot be attributed to cooling alone without controlled conditions.
Occupancy	Consider scenarios with different numbers of occupants.	Occupancy effects are not isolated from other room conditions.

Comparison Dimension	Purpose of Comparison	Interpretation Limit
Environmental parameters	Review temperature, humidity, light, and air-quality-related readings.	No validated causal or performance claim follows from descriptive values alone.

The scenario tables should be interpreted as descriptive project observations. They do not establish that occupancy or air conditioning alone caused the recorded differences, and they do not demonstrate that FreshAir Desk outperforms a reference monitor.

4.6 TESTING AND VERIFICATION

Verification Check	What to Observe / Record
Sensor acquisition	Check that each connected sensor field produces a readable value.

Verification Check	What to Observe / Record
Dashboard integration	**Confirm that the displayed fields correspond to the available sensor readings.**
Rule behavior	**Record a selected input condition and the resulting rule-based label.**
AI Insights consistency	**Compare generated text with the sensor values and dashboard status; note mismatches.**
Runtime observation	**Record missing, unstable, or changing readings during an observation period.**
MQ135 interpretation	**Ensure the reported values are not described as calibrated gas concentration without calibration evidence.**

The supplied material documents the system workflow and the dashboard functions but does not provide a complete controlled validation dataset, reference-instrument calibration results, or numerical accuracy metrics. Accordingly, results are described functionally rather than as verified sensor accuracy.

4.7 DASHBOARD OUTPUT

The figure below shows the FreshAir Desk dashboard running in a laptop browser. It displays the current temperature, humidity, light, and MQ135 air-quality readings with their condition labels and hints, the AI Insights panel, and a short history table.

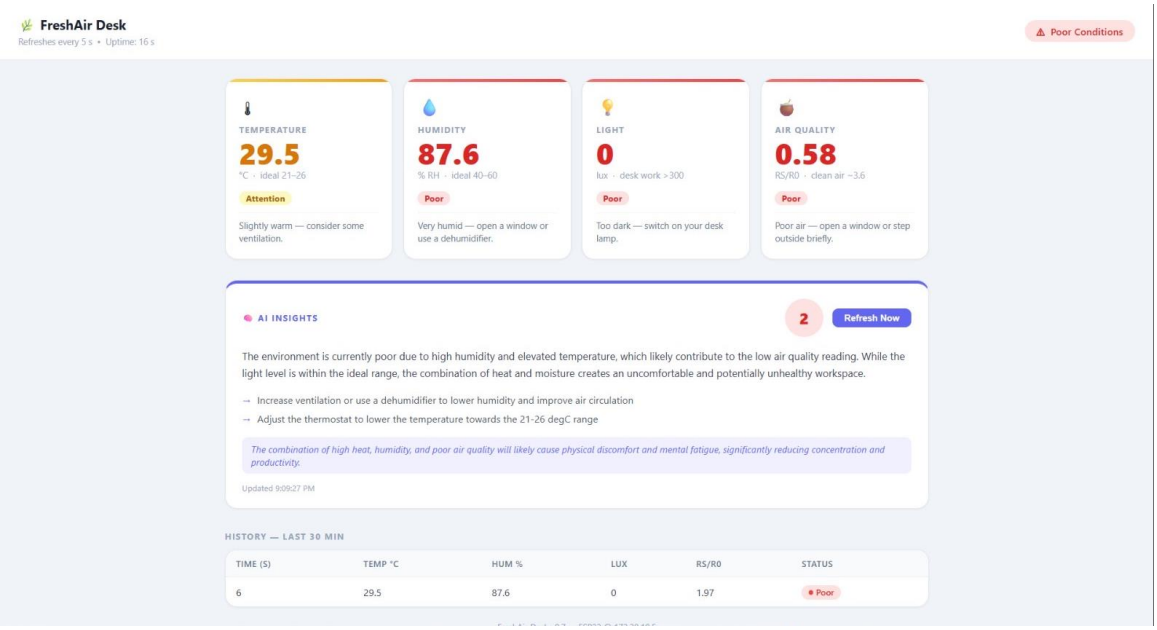


Figure 4.4: FreshAir Desk dashboard output

CHAPTER 5. CONCLUSION AND FUTURE WORK

5.1 CONCLUSION

FreshAir Desk was developed as an ESP32-based smart desk environment monitoring prototype that brings together temperature, relative humidity, ambient light intensity, and MQ135 gas-sensing readings. The DHT22, BH1750, and MQ135 sensors provide the input data, while the ESP32 supports acquisition and processing. A laptop dashboard presents live values, rule-based condition labels, fixed hints, and a short history.

The project also includes an AI Insights panel that generates a summary, practical suggestions, an overall numeric environment indicator, and a focus-impact statement from the sensor readings. This panel is advisory and remains subject to consistency refinements. The system's contribution is the integration of selected environmental readings and their interpretation into one accessible interface. The available project materials do not establish calibrated gas concentration, reference-level accuracy, or a measured improvement in health, comfort, or productivity.

5.2 LIMITATIONS

- The documented sensor set does not include a noise sensor.
- The MQ135 output is not documented as calibrated gas concentration in ppm.
- The complete classification thresholds and all rule predicates are not specified in the supplied materials.
- The sampling interval and ESP32-to-laptop communication protocol are not documented in the materials reviewed.
- The AI-generated score and focus-impact statement are not validated measurements of comfort, attention, or productivity.

- AI-generated text may occasionally disagree with dashboard status, and refinements are ongoing.
- The comparative scenarios are descriptive and do not establish causation or comparative accuracy against reference equipment.

5.3 FUTURE ENHANCEMENTS

- Document the exact sensor thresholds and classification rules used in the firmware.
- Calibrate the MQ135 module against suitable reference conditions before reporting gas concentrations.
- Validate temperature, humidity, and light readings against reference instruments and report the observed error.
- Add structured logging and exportable data to support longer observation periods.
- Improve the consistency checks between the dashboard labels and AI-generated insights.
- Evaluate the dashboard with users to understand clarity and usability, without assuming productivity improvements.
- Extend comparative testing using controlled room conditions and clearly recorded occupancy, ventilation, and measurement procedures.

REFERENCES

[1] Shriram et al., study on IoT-based thermal comfort and indoor air-quality monitoring in workspaces, as listed in the supplied FreshAir Desk literature survey.

[2] Praveenraja and Manitha, study on air and sound pollution monitoring with HVAC integration, as listed in the supplied literature survey.

[3] Gabriel and Auer, study applying machine-learning methods to indoor air-pollution estimation, as listed in the supplied literature survey.

[4] Yeom, Kim, and Yoon, study of productivity prediction using environmental conditions and physiological signals, as listed in the supplied literature survey.

[5] Cicceri et al., study using air-quality sensing for room headcount estimation, as listed in the supplied literature survey.

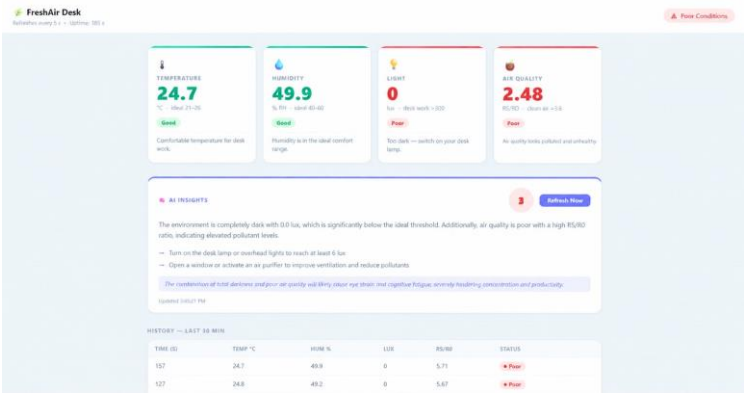
[6] Fanti et al., study on IoT-based smart-building management and occupant prediction, as listed in the supplied literature survey.

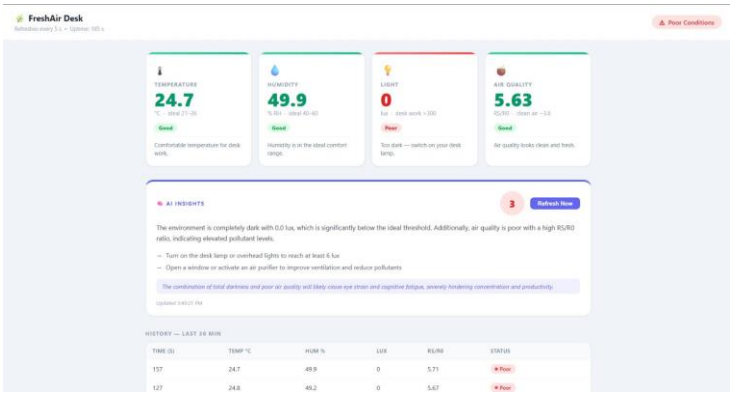
[7] Comparative study of intelligent and traditional buildings considering indoor environmental quality and self-reported productivity, as listed in the supplied literature survey.

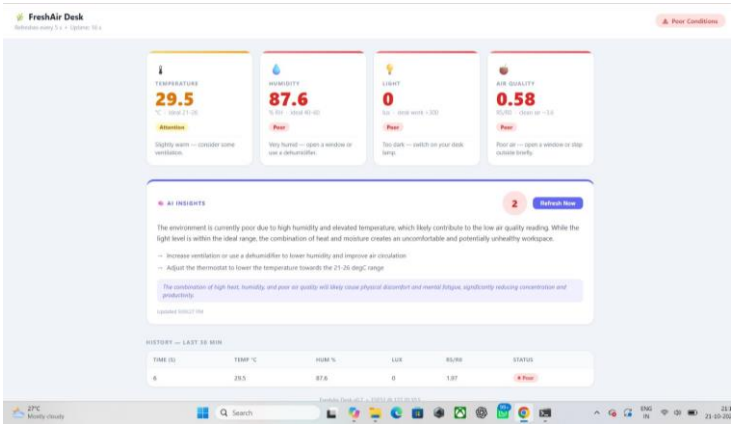
[8] M. S. Bakare and K. Abubaker, “IoT-based indoor environmental monitoring system using multi-parameter sensing and ESP32-WROOM integration,” Discover Electronics, vol. 3, article 6, 2026. doi: 10.1007/s44291-026-00157-3.

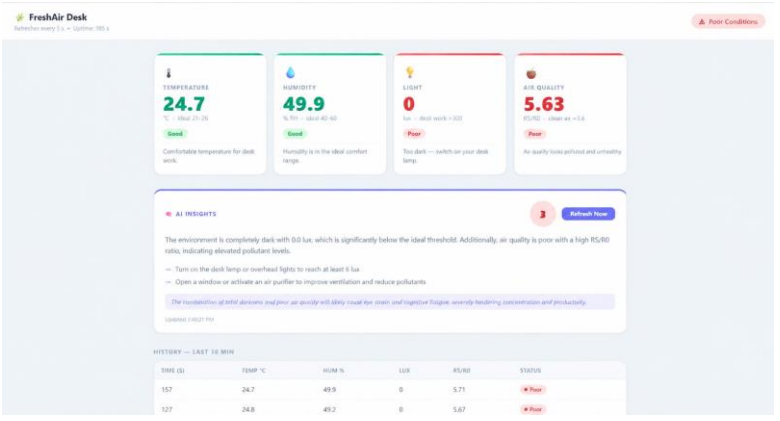
[9] [30] Additional indoor environmental quality, air monitoring, measurement, classroom, and lighting references are listed in the supplied FreshAir Desk literature survey. Full bibliographic details should be copied from that survey before final institutional submission.

SAMPLE INPUT AND OUTPUT









SOURCE CODE

```
import os, sys, time, json, threading, subprocess, requests
sys.stdout.reconfigure(encoding='utf-8', errors='replace')
from datetime import datetime
from flask import Flask, jsonify
from flask_cors import CORS
from dotenv import load_dotenv
from groq import Groq

load_dotenv()

api_key = os.environ.get("GROQ_API_KEY")
if not api_key:
    print("[ERROR] GROQ_API_KEY not set. Add it to ai_server/.env")
    exit(1)

client = Groq(api_key=api_key)

MODEL      = "qwen/qwen3.8-27b"
ESP32_URLS = ["http://192.168.137.100/json", "http://172.20.10.5/json"]
POLL_EVERY = 30   # seconds between automatic refreshes (Groq free: 14400 req/day)

app = Flask(__name__)
```

```
CORS(app)

cache = {
    "summary":      "Starting up — first reading in a moment...",
    "suggestions":  [],
    "score":        None,
    "focus_impact": "",
    "updated":      0,
    "error":        None
}
lock         = threading.Lock()
last_overall  = None   # track last known status for change detection


def now():
    return datetime.now().strftime("%H:%M:%S")


def build_prompt(data):
    return f"""You are an indoor desk environment assistant. Analyse the readings below and
respond with ONLY a JSON object — no markdown, no code fences.

Sensor readings right now:
- Temperature : {data.get('temp',  'N/A')} degC  (ideal 21-26 degC)
- Humidity    : {data.get('hum',   'N/A')} %     (ideal 40-60 %)
- Light       : {data.get('lux',   'N/A')} lux   (ideal 6 lux; <4 = too dark, >8 = too bright)
- Air quality : {data.get('ratio', 'N/A')} RS/R0 (clean air ~3.6; lower = more pollutants)
- Overall     : {data.get('overall', 'N/A')}

Return exactly this JSON:
{{
  "summary": "<2-3 sentences about current environment>",
  "suggestions": ["<action 1>", "<action 2>"],
  "score": <integer 1-10>,
  "focus_impact": "<one sentence on how this affects focus/productivity>"
}}"""


def parse_gemini_text(text):
    text = text.strip()
    if "```" in text:
        for part in text.split("```"):
            part = part.strip().lstrip("json").strip()
            try:
                return json.loads(part)
            except Exception:
                continue
    return json.loads(text)


def call_ai(data):
```

```
    prompt = build_prompt(data)
    for attempt in range(3):
        try:
            resp = client.chat.completions.create(
                model=MODEL,
                messages=[{"role": "user", "content": prompt}],
                temperature=0.3,
                max_tokens=400,
            )
            parsed = parse_gemini_text(resp.choices[0].message.content)
            print(f"[{now()}] Groq OK (attempt {attempt+1})", flush=True)
            return parsed
        except Exception as e:
            err_str = str(e)
            if "429" in err_str or "rate_limit" in err_str.lower():
                wait = 10 * (attempt + 1)
                print(f"[{now()}] Rate limited, waiting {wait}s...", flush=True)
                time.sleep(wait)
            elif "503" in err_str or "502" in err_str:
                wait = 5 * (attempt + 1)
                print(f"[{now()}] Groq overloaded, waiting {wait}s...", flush=True)
                time.sleep(wait)
            else:
                print(f"[{now()}] Groq error: {err_str[:120]}", flush=True)
                raise
    raise Exception("Groq failed after 3 attempts")


def fetch_and_update():
    global last_overall

    # Reach ESP32
    data = None
    for url in ESP32_URLS:
        try:
            resp = requests.get(url, timeout=5)
            resp.raise_for_status()
            data = resp.json()
            break
        except Exception:
            continue
    if data is None:
        msg = "ESP32 not reachable — check it is powered and on the same hotspot."
        print(f"[{now()}] {msg}")
        with lock:
            cache["error"] = msg
        return

    # Check if overall status changed — if so, force AI refresh regardless of timer
    current_overall = data.get("overall")
    changed = (current_overall != last_overall)
```

```
    if changed and last_overall is not None:
        print(f"[{now()}] Status changed: {last_overall} -> {current_overall}. Refreshing AI
suggestion...")
    last_overall = current_overall

    # Call Gemini
    try:
        parsed = call_ai(data)
        parsed["updated"] = int(time.time())
        parsed["error"]   = None
        with lock:
            cache.update(parsed)
        print(f"[{now()}] Suggestion updated. Score: {parsed.get('score')}/10 | {current_overall}")
    except Exception as e:
        print(f"[{now()}] All Gemini models failed: {e}")
        with lock:
            cache["error"] = "AI temporarily unavailable — will retry shortly."


def background_poller():
    global last_overall
    while True:
        time.sleep(POLL_EVERY)
        # Also force a refresh if status changed (poll ESP32 quickly to detect change)
        try:
            for url in ESP32_URLS:
                try:
                    data = requests.get(url, timeout=4).json()
                    current = data.get("overall")
                    if current != last_overall:
                        fetch_and_update()
                        break
                    break
                except Exception:
                    continue
            else:
                fetch_and_update()
        except Exception:
            fetch_and_update()


@app.route("/suggest")
def get_suggestion():
    with lock:
        return jsonify(cache)
@app.route("/suggest/refresh", methods=["POST"])
def refresh_now():
    # Fire in background — don't block the HTTP response
    threading.Thread(target=fetch_and_update, daemon=True).start()
    with lock:
        resp = dict(cache)
```

```
    resp["refreshing"] = True
    return jsonify(resp)
if __name__ == "__main__":
    all_ips = []
    try:
        out = subprocess.check_output("ipconfig", shell=True).decode(errors="ignore")
        for line in out.splitlines():
            if "IPv4" in line:
                ip = line.split(":")[-1].strip()
                if not ip.startswith("127."):
                    all_ips.append(ip)
    except Exception:
        pass
    # First fetch immediately in background
    threading.Thread(target=fetch_and_update, daemon=True).start()
    threading.Thread(target=background_poller, daemon=True).start()
    print()
    print("=" * 54)
    print("  FreshAir Desk -- AI Server")
    print("=" * 54)
    print("  Detected IPs:")
    for ip in all_ips:
        label = ""
        if ip.startswith("172.20.10."):   label = "  << iPhone hotspot"
        if ip.startswith("192.168.137."): label = "  << Rahul's PC hotspot (primary)"
        print(f"    {ip}{label}")
    print()
    print(f"  Model: {MODEL}  |  Poll every {POLL_EVERY}s")
    print("=" * 54)
    print()

    app.run(host="0.0.0.0", port=5000, debug=False)
```



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